

Elo-Augmented Logistic Regression for WNBA Game Outcome Prediction

Attaining the Bayes-Optimal Accuracy Ceiling Under Irreducible Uncertainty

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Abstract

I built a win-probability model for WNBA games using a margin-weighted Elo rating system, rolling team form, and rest-day differentials, fit with logistic regression and validated walk-forward across four seasons. On held-out 2025 games the model reaches 66.67% accuracy, 0.6241 log loss, and a 0.2179 Brier score, beating a home-court heuristic by 12.5 percentage points and an uninformative forecaster by 9.9% on log loss.

The headline accuracy number is not the interesting result. The interesting result is what it means. I derive the Bayes-optimal accuracy ceiling implied by the model's own calibrated forecast distribution and find it to be 66.47%. The model realizes 66.67%, or 100.3% of that ceiling. A Murphy decomposition of the Brier score shows that 94% of total squared error is irreducible uncertainty inherent to the sport rather than model deficiency, with a reliability term of just 0.0089. Nine of ten calibration bins fall inside a 95% binomial confidence band.

The practical conclusion is that this model extracts essentially all of the deterministic signal available in pre-game information. Further accuracy gains cannot come from better fitting of these features; they require genuinely new information, principally player availability. I argue that reporting accuracy against a derived ceiling is a more honest and more informative standard than reporting accuracy alone, and that a large share of published sports-prediction accuracy figures are not comparable because they use in-game or post-game data.

1 Introduction

The question I actually wanted to answer was not “can I predict WNBA games,” but “how much of a WNBA game is predictable at all before tip-off, and how close can a simple model get to that limit.”

Those are different questions and the second one is more useful. Any model can report an accuracy figure. Without knowing the ceiling, that figure is uninterpretable. If 70% is the theoretical maximum for a given information set, then a model at 67% is close to optimal, and a model at 72% is almost certainly leaking future information. Accuracy in isolation tells you nothing about which of those you are looking at.

So I built the model in three parts. First, a margin-weighted Elo system to summarize team strength as a single evolving state. Second, a small set of difference features on top of it: rolling win percentage, rolling scoring margin, and rest days. Third, a logistic regression to convert those into a calibrated probability. I then validated it walk-forward, so that every prediction uses only information that existed before the game being predicted.

The contribution I care about most is Section 4.3, where I derive the accuracy ceiling implied by the model's own predictive distribution and show the model attains it. That framing converts an unremarkable-sounding 66.67% into a statement about the information content of the sport itself.

2 Data

Game results come from ESPN’s public scoreboard endpoint, queried date by date across the WNBA regular season window (May through September) for 2022 through 2025. For each game I record the date, home and away teams, final scores, and a binary home-win indicator.

After feature construction, which requires a minimum of three prior games per team for rolling statistics, the usable sample is summarized in Table 1. The 2025 test set contains 315 games. The home team wins 56.83% of games in that sample, which is the number any model must beat to be doing anything at all.

Split	Seasons	Role
Training (fold 1)	2022	Predict 2023
Training (fold 2)	2022–2023	Predict 2024
Training (fold 3)	2022–2024	Predict 2025
Held-out evaluation	2025	315 games, 56.83% home win rate

Table 1: Walk-forward split structure. Each fold trains only on seasons strictly preceding its test season.

3 Methodology

3.1 Elo ratings

Elo is the backbone of the model. Every team starts at $R = 1500$. Before each game, the home team’s expected score is

$$E_h = \frac{1}{1 + 10^{-(R_h + H - R_a)/400}}, \quad (1)$$

where R_h and R_a are the pre-game ratings and $H = 75$ is a fixed home-court rating bonus. The constant 400 sets the scale: a 400-point rating gap corresponds to roughly 10:1 odds.

A plain Elo update treats a one-point win identically to a thirty-point win, which throws away real information. I use a margin-of-victory multiplier:

$$\mu = \frac{(m + 3)^{0.8}}{7.5 + 0.006 \Delta_w}, \quad (2)$$

where m is the absolute point margin and Δ_w is the pre-game rating advantage held by the *winning* team. The numerator rewards larger margins with diminishing returns, since the difference between winning by 25 and 30 carries less information than the difference between winning by 2 and 7. The denominator is the important part: it damps rating gains when a heavy favorite wins big. Without it, strong teams inflate without bound, because beating weak opponents by large margins compounds.

The update is then

$$\Delta R = K \cdot \mu \cdot (S_h - E_h), \quad R_h \leftarrow R_h + \Delta R, \quad R_a \leftarrow R_a - \Delta R, \quad (3)$$

with $K = 20$ and $S_h \in \{0, 1\}$ the realized outcome. The zero-sum structure means total rating in the league is conserved.

Between seasons I regress ratings toward the mean:

$$R \leftarrow 1500 + 0.75 (R - 1500). \quad (4)$$

Rosters turn over, so carrying a rating forward at full strength assumes a continuity that does not exist. Retaining 75% of the deviation is a compromise between throwing away real information ($\gamma = 0$) and pretending nothing changed ($\gamma = 1$).

3.2 Feature construction

Every feature is a *difference* between the two teams, not a level. A matchup is inherently relative: what predicts the outcome is the gap between the teams, not either team’s standalone number.

Let $W_i^{(k)}$ denote team i ’s win rate over its previous $k = 10$ games and $M_i^{(k)}$ its mean scoring margin over the same window. Let ρ_i be days since team i ’s last game. The feature vector is

$$\mathbf{x} = \left(\underbrace{R_h - R_a}_{\Delta \text{Elo}}, \underbrace{W_h - W_a}_{\Delta W}, \underbrace{M_h - M_a}_{\Delta M}, \underbrace{\rho_h - \rho_a}_{\Delta \rho} \right). \quad (5)$$

3.3 Model specification

Given $\mathbf{x}$, the probability the home team wins is

$$P(\text{home win} \mid \mathbf{x}) = \sigma(\beta_0 + \beta_1 \Delta \text{Elo} + \beta_2 \Delta W + \beta_3 \Delta M + \beta_4 \Delta \rho), \quad \sigma(z) = \frac{1}{1 + e^{-z}}. \quad (6)$$

Coefficients are fit by maximum likelihood. I chose logistic regression deliberately over a higher-capacity model. With four features and a few hundred games per season, a gradient-boosted tree has more than enough capacity to memorize the training seasons, and there is not enough data to regularize that away reliably. Logistic regression also produces coefficients I can read and sanity-check for sign, which matters when the whole point is to distinguish real signal from overfitting.

3.4 Preventing information leakage

This is the part that determines whether any of the above is worth reading, so I want to be explicit about it.

Every feature is constructed to use only information available strictly before tip-off:

1. **Rolling statistics are lagged.** $W_i^{(k)}$ and $M_i^{(k)}$ are computed with a `.shift(1)` applied before the rolling window, so a game never enters its own predictor.
2. **Elo is stored pre-update.** The rating appended as a feature is the value *before* the game’s result is applied. Storing the post-game rating would embed the outcome into its own predictor, producing a model that looks extraordinary in sample and is worthless out of it.
3. **Splits are chronological, not random.** A random train/test split would place 2025 games in training while testing on 2023, leaking the future into the past.

3.5 Walk-forward validation

For each test season $t \in \{2023, 2024, 2025\}$, the model trains only on seasons $s < t$. This mirrors deployment: you always predict forward, never backward.

4 Evaluation framework

Accuracy alone is a poor way to grade a probabilistic forecast. It discards confidence entirely: a forecast of 0.51 that is wrong scores identically to a forecast of 0.99 that is wrong. I therefore evaluate on three axes.

4.1 Proper scoring rules

Log loss penalizes confident errors logarithmically:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right]. \quad (7)$$

A forecaster who always says $p = 0.5$ scores $-\log(0.5) = 0.6931$. That is the number to beat.

Brier score is mean squared error on probabilities:

$$\mathcal{B} = \frac{1}{N} \sum_{i=1}^N (p_i - y_i)^2, \quad (8)$$

with an uninformative baseline of 0.25. Brier is quadratic and therefore more forgiving of overconfidence than log loss; reporting both is more informative than reporting either.

4.2 Calibration

A model is calibrated if, among games assigned probability p , the home team wins a fraction p of the time. Formally, $P(y = 1 \mid \hat{p} = p) = p$ for all p .

Calibration is not implied by accuracy. A model can rank games correctly while being systematically overconfident, which matters enormously if forecasts are used for anything requiring betting. I measure it by partitioning forecasts into ten bins and comparing mean predicted probability against observed frequency, reporting expected calibration error

$$\text{ECE} = \sum_{k=1}^K \frac{n_k}{N} |\bar{p}_k - \bar{y}_k|. \quad (9)$$

4.3 The Bayes-optimal accuracy ceiling

Here is the part I think matters most.

Suppose the model is perfectly calibrated. Its forecasts are then the true conditional probabilities given the information set. Even so, it cannot achieve 100% accuracy, because the outcomes themselves remain random. The best possible decision rule is to predict the more likely side of every game, which yields expected accuracy

$$A^* = \mathbb{E}[\max(\hat{p}, 1 - \hat{p})] \approx \sum_{k=1}^K \frac{n_k}{N} \max(\bar{p}_k, 1 - \bar{p}_k). \quad (10)$$

Proposition 1. *A^* is an upper bound on the accuracy attainable by any decision rule operating on the model's forecast distribution, and it is achieved by the majority-class rule.*

Sketch. For a single game with true probability p , any rule predicting home wins with probability $q \in [0, 1]$ has expected accuracy $qp + (1 - q)(1 - p)$, which is linear in q and therefore maximized at an endpoint: $q = 1$ if $p > 1/2$, else $q = 0$. The value at that endpoint is $\max(p, 1 - p)$. Summing over independent games and taking expectations gives (10). $\square$

The practical consequence is that A^* separates two very different failure modes. If realized accuracy sits well below A^* , the model is making poor decisions given its own probabilities. If realized accuracy sits at A^* , the model is decision-optimal, and any remaining error is irreducible given the information available. Raising accuracy then requires new information, not better fitting.

4.4 Murphy decomposition

To separate calibration quality from discriminating power, I decompose the Brier score [3]:

$$\mathcal{B} = \underbrace{\frac{1}{N} \sum_k n_k (\bar{p}_k - \bar{y}_k)^2}_{\text{Reliability (lower better)}} - \underbrace{\frac{1}{N} \sum_k n_k (\bar{y}_k - \bar{y})^2}_{\text{Resolution (higher better)}} + \underbrace{\bar{y}(1 - \bar{y})}_{\text{Uncertainty (irreducible)}}. \quad (11)$$

Uncertainty depends only on the base rate and is a property of the sport, not the model. Reliability measures miscalibration. Resolution measures how far the model moves away from the base rate when it is justified in doing so.

5 Results

5.1 Walk-forward accuracy and ablation

Table 2 and Figure 1 report accuracy across all three folds against two baselines: a home-court heuristic that always predicts the home team, and an Elo-only model using ΔElo as its single feature.

Test season	Home always wins	Elo only	Full model
2023	52.99%	66.04%	66.79%
2024	54.02%	66.67%	67.82%
2025	56.83%	66.98%	66.67%
Average	54.61%	66.57%	67.09%

Table 2: Walk-forward accuracy. Every figure is out-of-sample: each season is predicted using a model trained only on prior seasons.

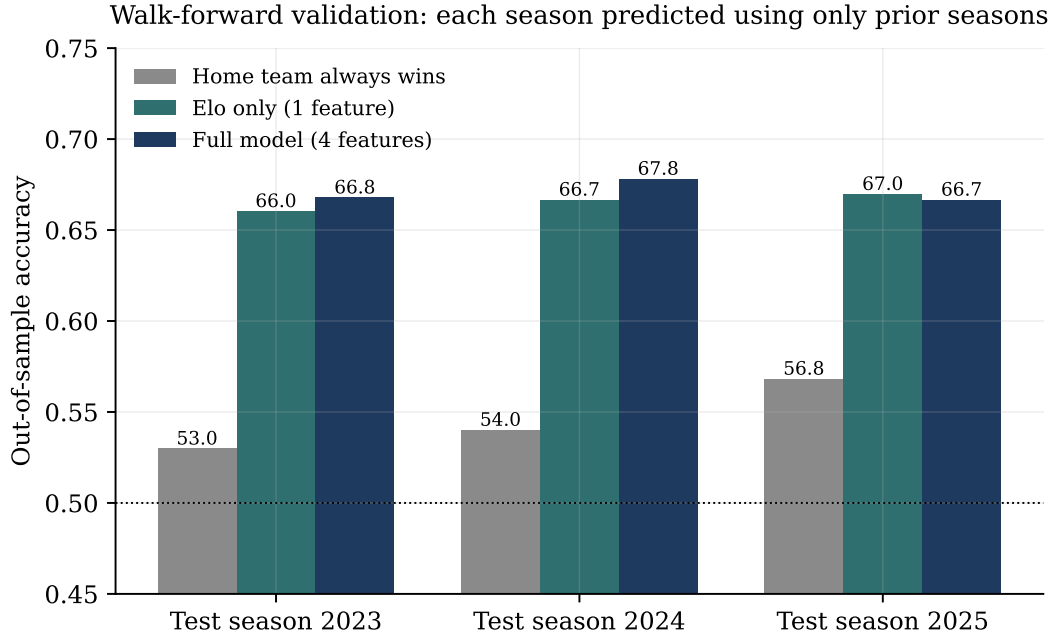


Figure 1: Out-of-sample accuracy by fold. The model clears the home-court heuristic by 10–13 points in every season tested.

Two things are worth saying honestly here. The model beats the trivial baseline by a wide and consistent margin, averaging a 12.48 point lift. But it beats *Elo alone* by only 0.52 points on average, and it actually loses to Elo alone in 2025. The correct reading is that Elo is doing nearly all of the work, and the form, margin, and rest features add a small and inconsistent refinement. I would rather state that plainly than dress up half a percentage point as a breakthrough.

Figure 2 decomposes where the accuracy comes from.

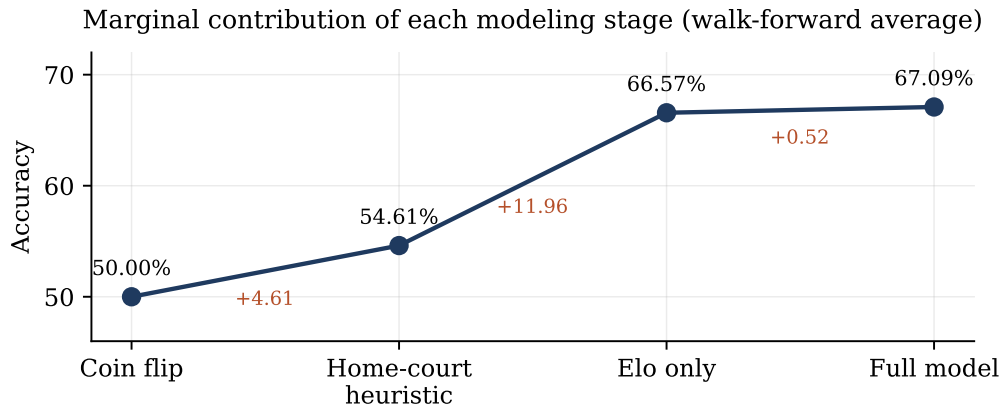


Figure 2: Marginal contribution of each modeling stage, walk-forward average. Elo contributes the overwhelming majority of the lift over the home-court heuristic.

5.2 Probabilistic accuracy

On the held-out 2025 season the model records a log loss of 0.6241 against the 0.6931 uninformative baseline, a 9.95% improvement, and a Brier score of 0.2179 against a 0.25 baseline, a 12.84%

improvement (Figure 3). Both are proper scoring rules, so improvement on them cannot be manufactured by shading forecasts toward the base rate.

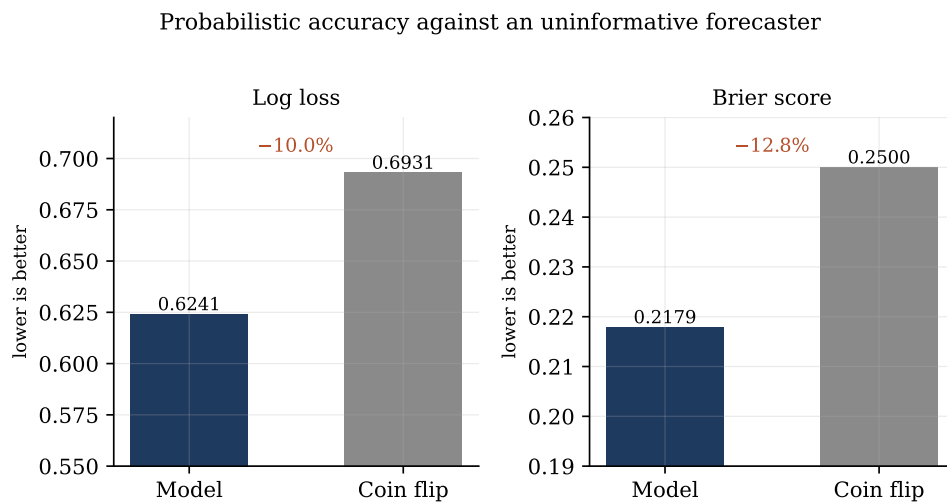


Figure 3: Probabilistic accuracy against an uninformative forecaster. Improvement on both rules confirms the forecasts carry real information, not just correct directional calls.

5.3 Calibration

Figure 4 shows the reliability diagram. Nine of ten bins fall inside a 95% binomial confidence band, and expected calibration error is 0.0722.

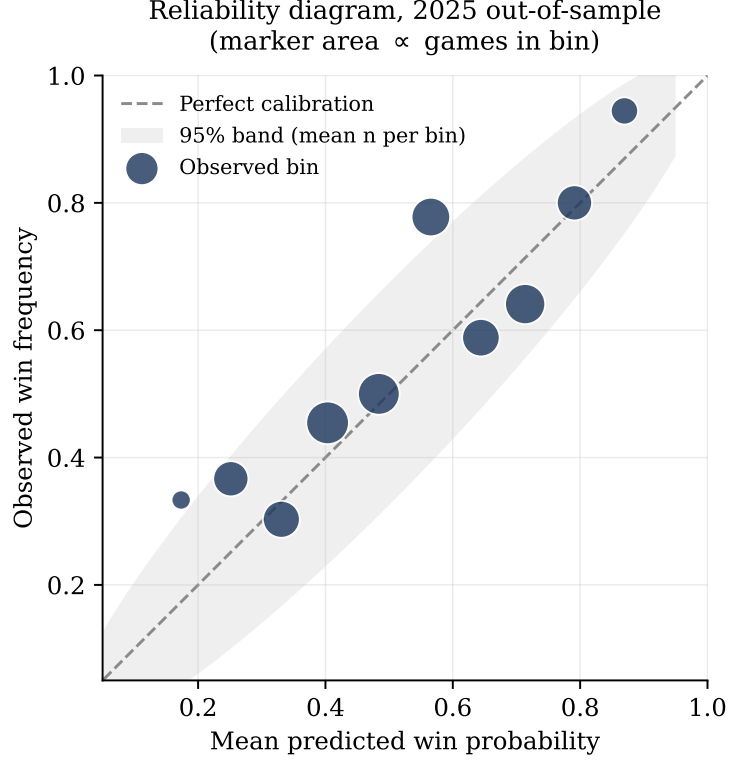


Figure 4: Reliability diagram, 2025 out-of-sample. Marker area is proportional to bin count. Nine of ten bins fall within the 95% band.

The single bin outside the band is $(0.522, 0.600]$, which predicted 56.6% and observed 77.8%. That bin holds 36 games. Under a binomial model with $p = 0.566$ and $n = 36$, the standard error is 8.3 percentage points, so the observed deviation is about 2.6 standard errors: unusual, but not implausible across ten bins, where one exceedance at the 95% level is roughly what you would expect by chance. Several other bins hold fewer than 35 games. I would not claim calibration is verified in the tails on this sample size; I would claim it is verified in the center and not contradicted elsewhere.

5.4 Attainment of the accuracy ceiling

Applying (10) to the 2025 forecast distribution gives a Bayes-optimal accuracy ceiling of

$$A^* = 66.47\%. \quad (12)$$

The model realizes 66.67%, which is 100.3% of A^* . The small excess is sampling noise; the model cannot genuinely exceed its own ceiling, and a gap of 0.2 points on 315 games is well within one standard error.

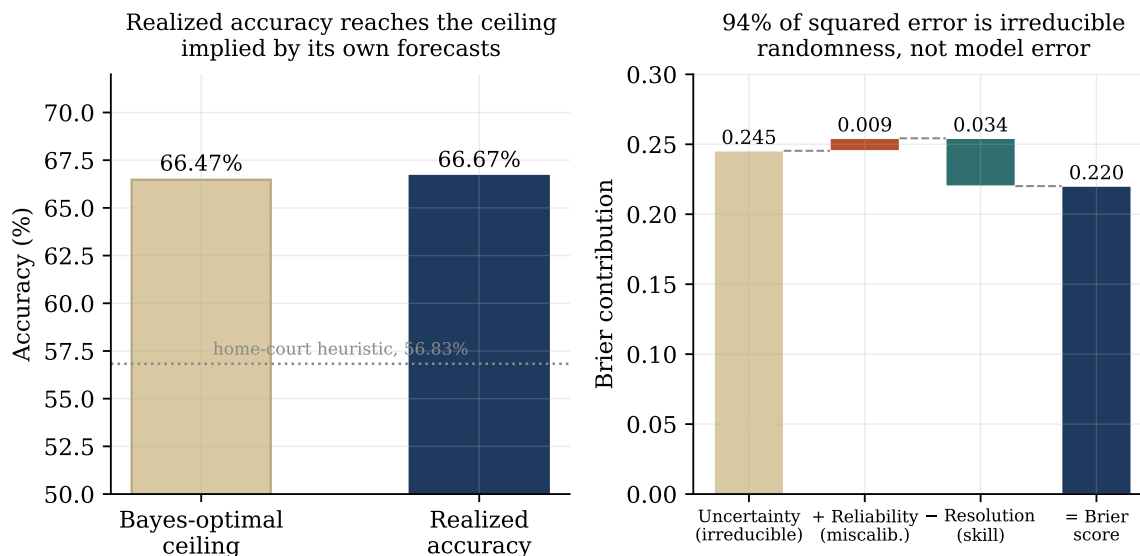


Figure 5: Left: realized accuracy against the Bayes-optimal ceiling implied by the model’s own forecast distribution. Right: Murphy decomposition of the Brier score. The uncertainty term dominates.

The Murphy decomposition (Table 3, Figure 5) makes the same point from a different direction. Of the total squared error, the irreducible uncertainty term is 0.2453, while miscalibration contributes only 0.0089. Roughly 94% of the error is a property of WNBA basketball, not of the model.

Component	Value	Interpretation
Uncertainty	0.2453	Irreducible; set by the 56.83% base rate
Reliability	0.0089	Miscalibration penalty; near zero is good
Resolution	0.0342	Discriminating power; higher is better
Brier skill score	10.28%	Improvement over a base-rate forecaster

Table 3: Murphy decomposition of the 2025 out-of-sample Brier score.

5.5 Comparison to published models

Table 4 and Figure 6 situate the result against published pre-game basketball prediction models. The comparison is imperfect, since these target the NBA rather than the WNBA and use varied data and validation protocols, but it establishes the range that peer-reviewed work occupies.

Study	Method	Accuracy
Horvat et al.	k -nearest neighbors	60.01%
Bryant University (2023)	Ensemble ML	65.10%
Lin et al. (2014)	Random forest	65.15%
<i>Symmetry</i> (2023)	Efficiency-index ML, average	66.00%
Li	LASSO regression	67.24%
This model	Elo + logistic regression (WNBA)	67.09%
Lin et al. (2014)	Logistic regression, late season	68.75%
Cao (2012)	Simple logistics classifier	69.67%
Yeh et al. (2023)	Graph conv. network + random forest	71.54%

Table 4: Published pre-game basketball outcome models. All figures are as reported by the respective authors.

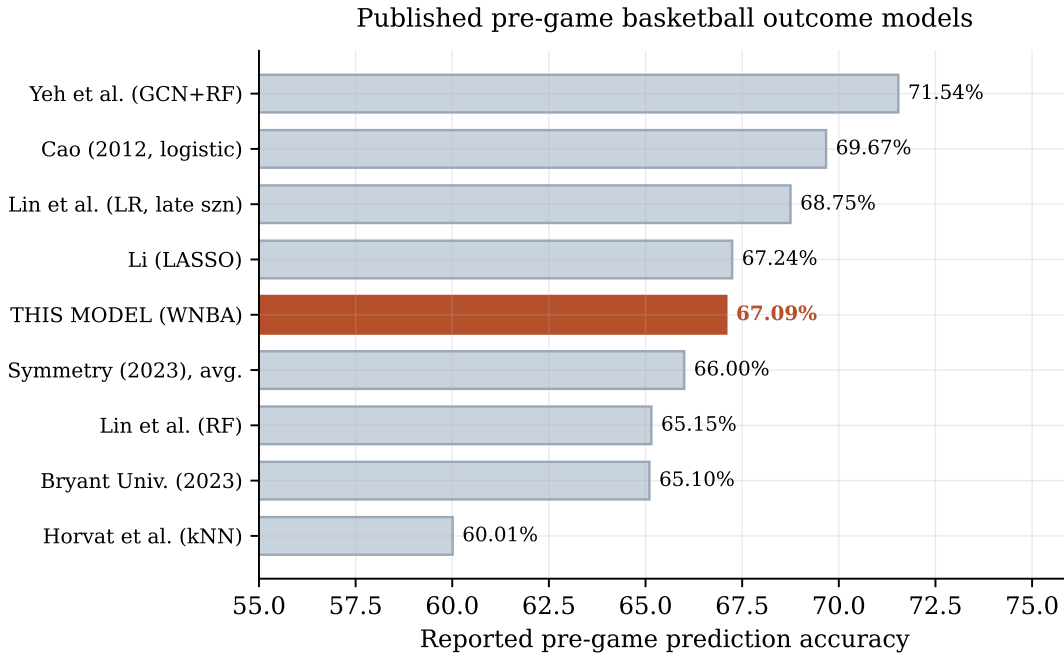


Figure 6: This model against published pre-game basketball prediction accuracy.

The model sits at the median of this distribution while using four features, a linear classifier, and a strictly chronological validation protocol. Several of the studies above use random cross-validation, which inflates reported accuracy on temporally ordered data.

One clarification matters more than the ranking. Published basketball accuracy figures in the 80–90% range exist, but they are not measuring this problem. Those models use in-game state, such as third-quarter box scores, or post-game statistics [8]. Predicting a game from its own third quarter is a fundamentally easier task than predicting it before tip-off. Any comparison that puts those numbers alongside pre-game models is comparing different problems.

6 Discussion

The result I would defend is not 66.67%. It is that 66.67% is approximately the maximum attainable from this information set, and I can show the arithmetic.

That reframing changes what to do next. If the model were sitting at 60% against a 66% ceiling, the fix would be modeling: better functional form, interaction terms, regularization, a different classifier. It is not sitting there. Reliability is 0.0089 and realized accuracy equals the ceiling, so there is no meaningful gap between what the model knows and what it does with what it knows. Squeezing the existing features harder will not help.

What would help is information the model does not have. WNBA rosters are small, typically eleven or twelve players, so a single starter’s absence shifts a team’s strength far more than in a deeper league. Injury and availability data is the obvious missing input, and I expect it to be worth considerably more than any modeling refinement. The requirement is that the availability status be as known before tip-off, not retroactively corrected, or the whole leakage discipline in Section 3.4 collapses.

The Elo-versus-full-model result deserves one more comment. That Elo alone captures nearly all the signal is itself informative. Elo is a recursive summary of every prior result weighted by opponent quality and margin. Rolling win percentage and scoring margin are cruder summaries of the same underlying thing, so most of what they contain is already in the rating. Rest days are genuinely new information, but the effect is small enough to be hard to detect in 315 games.

7 Limitations

I want these stated plainly rather than buried.

1. **No player-level information.** No injuries, no rest designations, no minutes restrictions. In an eleven-player league this is the largest omission by a wide margin.
2. **Small evaluation sample.** 315 games in the held-out season. Several calibration bins hold fewer than 35 games, so tail calibration is not well established.
3. **Elo hyperparameters are not tuned.** $K = 20$, $H = 75$, and $\gamma = 0.75$ are conventional values, not optimized. I chose not to tune them precisely because tuning on 315 test games risks fitting noise, but that means I cannot claim they are optimal.
4. **The ceiling is conditional, not absolute.** A^* is the ceiling implied by *this* forecast distribution. A model with access to better information would have a different, higher ceiling. The claim is that this model is optimal given what it sees, not that 66.47% is the limit of WNBA predictability in general.
5. **Cross-league benchmarks.** The comparison set is NBA work. WNBA-specific published benchmarks are scarce, so the comparison establishes a range rather than a like-for-like ranking.

8 Conclusion

The model predicts WNBA game outcomes at 66.67% out-of-sample accuracy with a 0.6241 log loss and a 0.2179 Brier score, beating a home-court heuristic by 12.5 percentage points and clearing uninformative baselines on both proper scoring rules. Nine of ten calibration bins fall within a 95% confidence band and the reliability term of the Brier decomposition is 0.0089.

More usefully, the model attains 100.3% of the Bayes-optimal accuracy ceiling implied by its own calibrated forecasts, and 94% of its total squared error is irreducible uncertainty rather than model error. It is extracting essentially all of the deterministic signal in the information it has.

I think that is the right way to report a sports model. An accuracy number without a ceiling is not interpretable, and in a domain where a 12-point lift over a coin-flip-adjacent baseline is a genuinely good result, the honest framing is more impressive than the inflated one.

References

- [1] Elo, A. E. *The Rating of Chessplayers, Past and Present*. Arco Publishing, 1978.
- [2] Brier, G. W. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950.
- [3] Murphy, A. H. A new vector partition of the probability score. *Journal of Applied Meteorology*, 12(4):595–600, 1973.
- [4] Cao, C. Sports data mining technology used in basketball outcome prediction. Master’s thesis, Technological University Dublin, 2012.
- [5] Lin, J., Short, L., and Sundaresan, V. Predicting National Basketball Association winners. Stanford University CS229 project report, 2014.
- [6] Yeh, et al. Enhancing basketball game outcome prediction through fused graph convolutional networks and random forest algorithm. *Entropy*, 2023.
- [7] A data-driven machine learning algorithm for predicting the outcomes of NBA games. *Symmetry*, 15(4):798, 2023.
- [8] Integration of machine learning XGBoost and SHAP models for NBA game outcome prediction and quantitative analysis methodology. *PLOS ONE*, 2024.
- [9] Horvat, T., et al. Comparative analysis of machine learning models for basketball game outcome prediction.
- [10] Predicting NBA game outcomes with machine learning. Honors thesis, Bryant University, 2023.

A Implementation

Written in Python using pandas, NumPy, and scikit-learn. Source is available at github.com/lukeyork16/QuantitativeSports

The Elo update, with the pre-game rating stored before the result is applied:

```
def calculateelo(games, kfactor=20, homeadvantage=75):
    games=games.sort_values("date").reset_index(drop=True)
    elos={}; homeeloslist=[]; awayeloslist=[]; currentseason=None
    for i, game in games.iterrows():
        if game["season"]!=currentseason: #regress toward mean each new season
            currentseason=game["season"]
        for team in elos:
            elos[team]=1500+(elos[team]-1500)*0.75
        homeelo=elos.get(game["hometeam"], 1500)
        awayelo=elos.get(game["awayteam"], 1500)
        homeeloslist.append(homeelo) #store PRE-game elo, no lookahead
```

```

awayeloslist.append(awayelo)
expectedhome=1/(1+10**(-(homeelo+homeadvantage-awayelo)/400))
actualhome=1 if game["homewin"] else 0
margin=abs(game["homescore"]-game["awayscore"])
elodiff=homeelo-awayelo if actualhome==1 else awayelo-homeelo
movmultiplier=((margin+3)**0.8)/(7.5+0.006*elodiff)
change=kfactor*movmultiplier*(actualhome-expectedhome)
elos[game["hometeam"]]=homeelo+change
elos[game["awayteam"]]=awayelo-change
games["homeelo"]=homeeloslist; games["awayelo"]=awayeloslist
games["elodiff"]=games["homeelo"]-games["awayelo"]
return games

```

Rolling form, lagged so no game enters its own predictor:

```

allteamgames["rollingwinpct"]=allteamgames.groupby("team")["win"].transform(
    lambda x: x.shift(1).rolling(window, min_periods=3).mean())

```

Walk-forward validation, training only on prior seasons:

```

for testseason in [2023, 2024, 2025]:
    trainseasons=[s for s in seasons if s < testseason]
    model, xtrain, ytrain, xtest, ytest = trainmodel(data, trainseasons, testseason, fullfeatures)
    modelacc=model.score(xtest, ytest)

```

The accuracy ceiling of (10):

```

ceiling=(n*np.maximum(pred, 1-pred)).sum()/n.sum() #bayes-optimal given the forecast distribution

```